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Introduction to Computer Network Ethics

Lecture 0.3 | CSE421 – Computer Networks

Department of Computer Science and Engineering
School of Data & Science

Objectives

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- Introduction to Morality
- Introduction to Ethics
- Ethics in Computer Networks

What is Morality

- Morality is a set of rules of right conduct, a system used to modify and regulate human behavior.
- A quality system to judge human acts right or wrong based on some moral standards.
- Morality is territorial and culturally based, also influenced by time.
- Moral codes are shared behavioral patterns of a group and responsible for the group’s success and failure.

What is Ethics

- Ethics is the study of right and wrong in human conduct.
  - Also, known as ‘theory of morals’
- Helps societies distinguish between right and wrong and gives a basis for justifying the judgment of human actions.
- The basis of trust and cooperation in relationships with others.

Ethical Theories

- Theories that help in justifying human action as good or bad are known as ethical theories.
- There are two types of ethical theories
  - Consequentialist Theories: focuses on the consequence of the action
    - Egoism
    - Utilitarianism
    - Altruism
  - Deontological Theories: focuses on the will of the action

Ethical Principles

- Ethical principles are tools which are used to think through difficult situations.
- Three useful ethical principles:
  - An act is ethical if all of society benefits from the act.
  - An act is ethical if people are treated as an end and not as a means to an end.
  - An act is ethical if it is fair to all parties involved.

Professional Codes of Ethics

- States the principles and core values that are essential to work of a particular occupational group.
- Helps in achieving,
  - Ethical decision making
  - High standards of practice and ethical behavior
  - Trust and respect from the general public
  - Evaluation benchmark
- Association for Computing Machinery's Code of Ethics can be found here: www.acm.org/about/code-of-ethics

Computer Ethics

- Four primary issues,
  - Privacy – responsibility to protect data about individuals
  - Accuracy - responsibility of data collectors to authenticate information and ensure its accuracy
  - Property - who owns information and software and how can they be sold and exchanged
  - Access - responsibility of data collectors to control access and determine what information a person has the right to obtain about others and how the information can be used

Computer Ethics for Computer Professionals

- **Competence**– Professionals keep up with the latest knowledge in their field and perform services only in their area of competence.
- **Responsibility**– Professionals are loyal to their clients or employees, and they won't disclose confidential information.
- **Integrity**– Professionals express their opinions based on facts, and they are impartial in their judgments.

Private Networks

- Employers may legally monitor electronic mail
  - In 2001, 63% of companies monitored employee Internet connections including about two-thirds of the 60 billion electronic messages sent by 40 million e-mail users.
- Most online services reserve the right to censor content
- These rights lead to contentious issues over property rights versus free speech and privacy

The Internet and the Web

- Most people don’t worry about email privacy on the Web due to *illusion of anonymity*
    - Each e-mail you send results in at least 3 or 4 copies being stored on different computers.
- Web sites often load files on your computer called *cookies* to record times and pages visited and other personal information
- ***Spyware -*** software that tracks your online movements, mines the information stored on your computer, or uses your computer for some task you know nothing about.

Internet Content & Free Speech Issues

- Information on internet includes hate, violence, and information that is harmful for children
  - How much of this should be regulated?
  - Do filters solve problems or create more?
- Is web site information used for course work and research **reliable**?

Why Ethics Matter in Networking

- Unethical behavior can harm individuals, organizations, and society (e.g., privacy breaches, cyberbullying, identity theft).
- Example: A student shares a friend’s password—later, sensitive information gets leaked and trust is broken.
- Acting ethically online ensures trust and safety for everyone.

Reasons for Internet Vulnerability

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- Sharing cloud resources, passwords and other authentication information.
- Computer Crimes
- Increased reliance on commercial software with known vulnerabilities.
    - Makes the user vulnerable to exploits.

Computer Crime

- Computer criminals -using a computer to commit an illegal act
- Who are computer criminals?
  - Employees – disgruntled or dishonest --the largest category
  - Outside users - customers or suppliers
  - “Hackers” and “crackers” - hackers do it “for fun” but crackers have malicious intent
  - Organized crime - tracking illegal enterprises, forgery, counterfeiting

Types of Exploits

- Viruses: piece of programming code in disguised that causes a computer to behave in an unexpected manner.
  - Spreads by the action of the “infected” computer user
- Worms: same as virus but doesn’t need user intervention to spread.
- Trojan Horses: a program where malicious code is hidden inside a seemingly harmless program.
- Other types: Spam, DDoS Attacks, Rootkits, Phishing, Smishing and Vishing

Computer Security

- Computer security involves protecting:
  - information, hardware and software
  - from unauthorized use and damage and
  - from sabotage and natural disasters

How to Ensure Ethics in Networking

- Implementing trustworthy computing
- Responsible network usage
- Maintaining privacy and confidentiality
- Ensuring strong passwords & authentication
- Avoiding malware & phishing
- Respecting intellectual properties
- Promoting social engineering awareness
- Following rules and regulations of the community
- Reporting and responding to incidents

Implementing Trustworthy Computing

- Microsoft maintains four pillars (security, privacy, reliability, business integrity) to ensure trustworthy computing
- Risk Assessment
- Establishing a security policy
- Educating Employees and Contract Workers
- Prevention
    - Installing corporate firewalls, antivirus software
    - Implementing safeguards
- Detection and Response

Responsible Network Usage

- Use network resources (internet, lab computers) for their intended purpose—usually learning and research.
- Avoid activities that hog bandwidth or disrupt others (e.g., streaming, mass downloading, online gaming during class).
- Don’t try to bypass school filters or security systems; this is considered misuse and may be against school policy.

Privacy and Confidentiality

- Always protect your own and others' information.
- Do not peek at, intercept, or share someone else's communications or data.
- Never post private details (like addresses or passwords) online, even if you think it's "just among friends."
- Think: Would I want this information about me shared?

Passwords and Authentication

- Use unique, strong passwords—combine letters, numbers, and symbols.
- Never share your password, even with close friends.
- Change your password if you suspect someone knows it.
- Real-life example: Weak passwords enabled a student to access classmates' assignments. The result: lost grades and disciplinary action.

Avoiding Malware & Phishing

- Watch out for unexpected emails or messages that encourage you to click links or download attachments.
- Don’t trust messages from unknown senders asking for personal info or money.
- If you’re unsure, delete the message or ask IT for help.
- Tip: “If in doubt, throw it out.”

Intellectual Property

- Respect copyrights—don’t share or download pirated movies, music, or software.
- Understand software licenses (some are free for student or educational use, others aren’t).
- Using pirated content can get you and your institution into legal trouble, and it’s unfair to creators.
- Avoid plagiarism of any means by copying text from other sources when original work is expected

Terms

- **INTELLECTUAL PROPERTY:**

  Intangible creations protected by law

- **TRADE SECRET:**

  Intellectual work or products belonging to a business, not in public domain

- **COPYRIGHT:**

  Statutory grant protecting intellectual property from copying by others for 28 years

- **PATENT:**

  Legal document granting owner exclusive monopoly on an invention for 17 years

Copyright Laws

- Software developers (or the companies they work for) own their programs.
- Software buyers only own the right to use the software according to the license agreement.
- No copying, reselling, lending, renting, leasing, or distributing is legal without the software owner's permission.

Software Licenses

- There are four types of software licenses:
  - Public Domain
  - Freeware
  - Shareware
  - All Rights Reserved

Public Domain License

- Public domain software has no owner and is not protected by copyright law.
- It was either created with public funds, or the ownership was forfeited by the creator.
- Can be copied, sold, and/or modified
- Often is of poor quality/unreliable

Freeware License

- Freeware is copyrighted software that is licensed to be copied and distributed without charge.
- Freeware is free, but it’s still under the owner’s control.
- Examples:
  - Eudora Light
  - Netscape

Shareware License

- A shareware software license allows you to use the software for a trial period, but you must pay a registration fee to the owner for permanent use.
  - Some shareware trials expire on a certain date
  - Payment depends on the honor system
- Purchasing (the right to use) the software may also get you a version with more powerful features and published documentation.

All Rights Reserved License

- May be used by the purchaser according the exact details spelled out in the license agreement.
- You can’t legally use it--or even possess it-- without the owner’s permission.

Software Piracy

- SPA (Software Publishers Association) polices software piracy and mainly targets:
  - Illegal duplication
  - Sale of copyrighted software
  - Companies that purchase single copies and load the software on multiple computers or networks
- They rely on whistle-blowers.
- Penalties (for primary user of PC) may include fines up to $250,000 and/or imprisonment up to 5 years in jail

Social Engineering Awareness

- Social engineering: tricks that manipulate you into giving up private information (e.g., phishing, fake tech support).
- Don’t give out your details to people you don’t know—even if they sound official.
- If something feels “off,” report it to your teacher or IT support.
- Example tactic: An email pretending to be from your school, asking you to “verify your password.”

Following Rules and Policies

- Many countries have laws protecting online data and punishing hacking (e.g., Computer Misuse Act).
- Brac University has some Code of Conducts that need to be followed as a member of the community.
- Take time to read and understand the rules and policies.

Safe Social Media & Online Behavior

- Don’t post things you wouldn’t want everyone (teachers, employers, family) to see.
- Don’t bully, harass, or “troll” others—this can have severe consequences.
- Set your privacy controls and think twice before sharing personal updates or photos.
- Follow Email Netiquettes.

E-Mail Netiquette

- Promptly respond to messages.
- Delete messages after you read them if you don’t need to save the information.
- Don’t send messages you wouldn’t want others to read.
- Keep the message short and to the point.
- Don’t type in all capital letters.
- Be careful with sarcasm and humor in your message.

Reporting & Responding to Incidents

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- If you experience or witness a security problem (hacking, bullying, data leak), report it to a teacher or IT support immediately.
- Quick action can prevent further damage.
- Reporting is not “snitching”—it’s about protecting yourself and your community.

Summary & Key Takeaways

- Ethics apply everywhere, including online.
- Treat others’ data and privacy as you’d want yours treated.
- Follow your school’s rules and ask when unsure.
- Reporting issues keeps your network and peers safe.
- Good online ethics build trust and help your reputation—now and in the future.

Reading

- Books:
  1. “Computer Network Security and Cyber Ethics” by Joseph Migga Kizza
     - Focused on chapter 2 & 3
  2. “Ethics in Information Technology” by George Reynolds
     - Focused on chapter 2 & 3